

Introduction

As a **non-profit organization** committed to **sustainable development**, we have decided to carry out an **analysis** aimed at examining **key aspects**, with the main goal of identifying which **countries** are more responsive to sustainable development and which, instead, require **immediate support** in terms of **health**, **education**, and **environmental impact**.

In particular, we aim to answer three **fundamental questions**:

- Where would investments in **renewable energy** have the **greatest impact**?
- Which countries show the most **critical social indicators** (especially in **education** and **health**)?
- Are there countries with strong **economic potential** but evident **energy inefficiencies**?

The analyses we present are intended both for our **donors**, who will be able to understand and evaluate the **strategic choices** behind the incurred costs, and for our **social media followers**. Sharing them publicly allows us to **raise awareness** about sustainability issues and, at the same time, promote a more **responsible lifestyle** in our own country.

Furthermore, this work contributes to strengthening our future **fundraising campaigns**, making them more **transparent** and **effective**.

Table Formatting

-- STEP 1: TABLE CREATION

```
CREATE TABLE global_country_1 (  
  Country          VARCHAR(55),  
  Density          DECIMAL(10,2),  
  Abbreviation     VARCHAR(10),  
  Agricultural_Land_percentual DECIMAL(4,3),  
  Land_Area        DECIMAL(10,2),  
  Armed_Forces_size DECIMAL(10,2),  
  Birth_Rate       DECIMAL(10,2),  
  Calling_Code     INT,  
  Capital          VARCHAR(55),  
  Co2_Emissions    DECIMAL(10,2),  
  CPI              DECIMAL(10,2),  
  CPI_Change_percentual DECIMAL(4,3),  
  Currency         VARCHAR(10),  
  Fertility_Rate   DECIMAL(4,3),  
  Forested_Area_percentual DECIMAL(4,3),  
  Gasoline_Price_dollar DECIMAL(4,3),  
  GDP_dollar       DECIMAL(20,3),  
  Gross_primary_education_enrollment DECIMAL(4,3),  
  Gross_tertiary_education_enrollment DECIMAL(4,3),  
  Infant_mortality DECIMAL(5,2),  
  Largest_city     VARCHAR(55),  
  Life_expectancy  DECIMAL(5,2),  
  Maternal_mortality_ratio INT,  
  Minimum_wage     DECIMAL(4,2),  
  Official_language VARCHAR(55),  
  Health_expenditure DECIMAL(4,3),  
  Physicians_per_thousand DECIMAL(4,3),  
  Population       DECIMAL(15,2),  
  Population_Labor_force_percentual DECIMAL(4,3),  
  Tax_revenue_percentual DECIMAL(4,3),  
  Total_tax_rate   DECIMAL(4,3),  
  Unemployment_rate DECIMAL(4,3),  
  Urban_population DECIMAL(15,2),  
  Latitude         DECIMAL(9,6),  
  Longitude        DECIMAL(9,6)  
);  
  
CREATE TABLE sustainable_energy (  
  Entity          VARCHAR(55),  
  Year            INT,  
  Access_to_electricity_percent DECIMAL(10,2),  
  Access_to_cooking_fuel DECIMAL(10,2),  
  Renewable_electricity_generating_capacity_capita DECIMAL(10,2),  
  Financial_flows_to_developing_countries BIGINT,
```

```

Renewable_energy_percent          DECIMAL(10,2),
Electricity_from_fossil_fuels      DECIMAL(10,2),
Electricity_from_nuclear           DECIMAL(10,2),
Electricity_from_renewables        DECIMAL(10,2),
Low_carbon_electricity_percent     DECIMAL(10,2),
Primary_energy_consumption_per_capita DECIMAL(10,2),
Energy_intensity_level_of_primary_energy DECIMAL(10,2),
Value_co2_emissions_kt_by_country DECIMAL(10,2),
Renewables_in_percent_as_primary_energy DECIMAL(10,2),
gdp_growth                        DECIMAL(10,2),
gdp_per_capita                    DECIMAL(10,2),
Density_n                         INT,
Land_Area_in_km2                  DECIMAL(12,5),
Latitude                          DECIMAL(9,6),
Longitude                         DECIMAL(9,6)
);

```

-- STEP 2: LINKING THE TWO TABLES

```

ALTER TABLE global_country_1
ADD COLUMN id SERIAL PRIMARY KEY;

```

```

ALTER TABLE sustainable_energy
ADD COLUMN country_id INT;

```

```

UPDATE sustainable_energy
SET country_id = global_country_1.id
FROM global_country_1
WHERE sustainable_energy.entity = global_country_1.country;

```

```

-- Check for countries that don't match between the two tables
SELECT DISTINCT entity
FROM sustainable_energy
WHERE entity NOT IN (SELECT country FROM global_country_1);

```

```

ALTER TABLE sustainable_energy
ADD CONSTRAINT fk_country
FOREIGN KEY (country_id) REFERENCES global_country_1(id);

```

```

ALTER TABLE global_country_1
RENAME TO global_country;

```

```

SELECT * FROM global_country;

```

First Question

QUESTION 1: Where would investments in renewable energy have the greatest impact?

The analysis focuses on identifying countries where investments in **renewable energy** could generate the highest **environmental and social impact**.

Several key factors are considered. A high **dependence on non-renewable sources** indicates a greater opportunity for transformation, as the introduction of renewables can significantly reduce environmental impact. **Access to electricity** is another crucial element, since countries with limited access or weak infrastructure would benefit the most from targeted interventions.

Additionally, **CO2 emissions** are taken into account, as introducing clean energy in high-emission contexts can strongly reduce **pollution** and improve overall environmental conditions. The **fertility rate** is also relevant from a long-term perspective, as younger populations may offer greater potential for **awareness** and **behavioral change**.

Finally, **population size** and **population density** are considered, prioritizing highly populated countries to maximize impact, or densely populated areas where even small interventions can produce significant results.

```
WITH renewable_context AS (
  SELECT
    s.entity,
    ROUND(AVG(s.renewable_energy_percent), 2) AS avg_ren_elect_last_3_years
  FROM sustainable_energy s
  WHERE s.year BETWEEN 2017 AND 2019
    AND s.renewable_energy_percent IS NOT NULL
  GROUP BY s.entity
),
social_indicators AS (
  SELECT
    r.entity,
    r.avg_ren_elect_last_3_years,
    s.access_to_electricity_percent,
    g.co2_emissions,
    g.fertility_rate,
    s.density_n,
    g.population
  FROM renewable_context r
  JOIN sustainable_energy s ON r.entity = s.entity
  JOIN global_country g ON s.country_id = g.id
  WHERE s.year = 2019
    -- Low access to electricity (below world average)
    AND s.access_to_electricity_percent < (
```

```

        SELECT AVG(access_to_electricity_percent) FROM sustainable_energy
    )
    -- High fertility rate (above world average)
    AND g.fertility_rate > (
        SELECT AVG(fertility_rate) FROM global_country
    )
    -- Large population OR high density
    AND (
        g.population > (SELECT AVG(population) FROM global_country)
        OR
        s.density_n > (SELECT AVG(density_n) FROM sustainable_energy)
    )
)
SELECT *
FROM social_indicators
-- CO2 emissions used as primary sort: highest emitters first,
-- then lowest renewable share to identify the biggest opportunities
ORDER BY
    co2_emissions DESC,
    avg_ren_elect_last_3_years ASC
LIMIT 5;

```


Second Question

QUESTION 2: Which countries show the most critical social indicators (particularly education and health)?

The goal is not to identify the “worst” countries in absolute terms, but to find those where a **social intervention** could have the greatest impact on **human wellbeing**.

The analysis focuses on several key factors. **Life expectancy** is considered a primary indicator, as low values signal an urgent need for improvements in **health conditions**. The **education level** is another crucial element, since education represents a fundamental driver of **long-term sustainable development**.

In addition, **access to electricity** plays a critical role, as the lack of it significantly reduces the effectiveness of both **educational** and **health interventions**. Finally, the **fertility rate** is taken into account: when high and combined with low levels of education and health, it can represent a major **structural challenge** for a country’s development.

WITH renewable_context AS (

SELECT s.entity, ROUND(AVG(s.renewable_energy_percent), 2) AS
avg_renewable_last_years

FROM sustainable_energy s

WHERE s.year BETWEEN 2017 AND 2019

AND s.renewable_energy_percent IS NOT NULL

GROUP BY s.entity

),

social_indicators AS (

SELECT s.entity, g.life_expectancy, g.gross_tertiary_education_enrollment,
s.access_to_electricity_percent, g.fertility_rate, g.population

FROM sustainable_energy s

JOIN global_country g ON s.country_id = g.id

WHERE s.year = 2019

)

```

SELECT

    r.entity, r.avg_renewable_last_years, si.life_expectancy,
    si.gross_tertiary_education_enrollment, si.access_to_electricity_percent, si.fertility_rate,
    si.population

FROM renewable_context r

JOIN social_indicators si ON r.entity = si.entity

WHERE

    -- Salute critica

    si.life_expectancy < (SELECT AVG(life_expectancy) FROM global_country)

    -- Istruzione bassa

    AND si.gross_tertiary_education_enrollment <

        (SELECT AVG(gross_tertiary_education_enrollment) FROM global_country)

    -- Accesso all'elettricità limitato

    AND si.access_to_electricity_percent <

        (SELECT AVG(access_to_electricity_percent) FROM sustainable_energy)

    -- Fertilità alta

    AND si.fertility_rate >

        (SELECT AVG(fertility_rate) FROM global_country)

ORDER BY

    si.life_expectancy ASC,

    si.gross_tertiary_education_enrollment ASC

LIMIT 5;

```

Third Question

QUESTION 3: Are there countries with strong economic potential but clear energy inefficiencies?

In this section, the focus shifts from the most vulnerable countries to those that already possess many of the conditions needed to make a significant leap toward sustainability. The objective is to maximize the environmental and social impact of interventions by developing pilot projects that can later be replicated in other contexts.

Countries with strong economic potential are identified through several key indicators. A **GDP per capita above the global average** suggests a solid economic foundation capable of supporting investments and structural reforms. Additionally, **GDP growth over time** reflects dynamic economies where energy efficiency initiatives can be integrated into ongoing development processes. A **high level of access to electricity** further indicates the presence of essential infrastructure, allowing for improvements without the need to build systems from scratch.

On the other hand, energy inefficiency is primarily highlighted by a **low share of renewable energy**, which points to significant room for improvement in the energy transition.

WITH min_max_year AS (

SELECT

entity,

year,

gdp_per_capita

FROM sustainable_energy

WHERE year IN (2009, 2019)

),

differenza AS (

SELECT

```

entity,

MAX(CASE WHEN year = 2009 THEN gdp_per_capita END) AS gdp_2009,

MAX(CASE WHEN year = 2019 THEN gdp_per_capita END) AS gdp_2019

FROM min_max_year

GROUP BY entity

HAVING COUNT(*) = 2

),

growth AS (

SELECT

d.entity,

d.gdp_2009,

d.gdp_2019,

ROUND(((d.gdp_2019 - d.gdp_2009) / d.gdp_2009) * 100, 2) AS growth_percent,

s.access_to_electricity_percent,

s.renewable_energy_percent

FROM differenza d

JOIN sustainable_energy s ON d.entity = s.entity

AND s.year = 2019

WHERE d.gdp_2009 IS NOT NULL

AND d.gdp_2019 IS NOT NULL

AND d.gdp_2019 > (SELECT AVG(gdp_2019) FROM differenza)

AND s.access_to_electricity_percent > (SELECT AVG(access_to_electricity_percent)

FROM sustainable_energy)

AND s.renewable_energy_percent < (SELECT AVG(renewable_energy_percent) FROM

sustainable_energy)

)

SELECT *

```

FROM growth

ORDER BY growth_percent DESC

LIMIT 10;

/*

A second query is run with the same structure but inverting the renewable energy filter: here we select countries ABOVE the global average in renewable share. These are the "model countries" (such as Uruguay, Lithuania, and Iceland) that have already successfully transitioned and can serve as inspiration.

We also add tertiary education enrollment to highlight the connection between education and renewable energy adoption.

*/

WITH min_max_year AS (

SELECT

entity,

year,

gdp_per_capita

FROM sustainable_energy

WHERE year IN (2009, 2019)

),

differenza AS (

SELECT

entity,

MAX(CASE WHEN year = 2009 THEN gdp_per_capita END) AS gdp_2009,

```

MAX(CASE WHEN year = 2019 THEN gdp_per_capita END) AS gdp_2019

FROM min_max_year

GROUP BY entity

HAVING COUNT(*) = 2

),

growth AS (

SELECT

    d.entity,

    d.gdp_2009,

    d.gdp_2019,

    ROUND(((d.gdp_2019 - d.gdp_2009) / d.gdp_2009) * 100, 2) AS growth_percent,

    s.access_to_electricity_percent,

    s.renewable_energy_percent,

    g.gross_tertiary_education_enrollment

FROM differenza d

JOIN sustainable_energy s ON d.entity = s.entity

JOIN global_country g ON g.country = d.entity

    AND s.year = 2019

WHERE d.gdp_2009 IS NOT NULL

    AND d.gdp_2019 IS NOT NULL

    AND d.gdp_2019 > (SELECT AVG(gdp_2019) FROM differenza)

    AND s.access_to_electricity_percent > (SELECT AVG(access_to_electricity_percent)
FROM sustainable_energy)

    AND s.renewable_energy_percent > (SELECT AVG(renewable_energy_percent) FROM
sustainable_energy)

)

SELECT *

```


FROM growth

ORDER BY growth_percent DESC

LIMIT 5;

Insights and Strategic Focus

In this section, the focus shifts from the most vulnerable countries to those that already possess many of the conditions needed to make a significant leap toward sustainability. The objective is to maximize the environmental and social impact of interventions by developing pilot projects that can later be replicated in other contexts.

Countries such as **Singapore**, the **United States**, and **Malta** represent significant untapped potential. These are economies with strong foundations and well-developed infrastructure, yet they still exhibit energy inefficiencies and lower-than-expected adoption of renewable energy.

In contrast, countries like **Uruguay**, **Lithuania**, and **Iceland** stand out as positive examples. They have successfully advanced in the energy transition, demonstrating that meaningful results can be achieved through targeted and consistent policies. However, differences in socio-cultural, environmental, and territorial contexts make it challenging to directly replicate these models in other regions.

For this reason, and considering the relatively smaller scale of these “model countries,” the focus shifts toward contexts such as **Singapore** and **Malta**, where high-impact experimental projects may be easier to implement and later adapted to similar environments.

Finally, countries with higher levels of renewable energy adoption also tend to show above-average **tertiary education enrollment** (global average: ~0.37), further highlighting the crucial role of **education** in driving sustainable development.